

Contents

1 、 Introduction	3
2 、 User Information.....	4
3 、 Basic Concepts	4
3.1. BASIC CONCEPTS	4
3.1.1. User Enrollment	4
3.1.2. User Verification	4
3.1.3. Match Threshold Levels	5
3.1.4. User ID numbers	5
3.1.5 Authority(status) Levels status.	5
3.1.6. Start-up window.....	6
3.2. HOW TO PLACE THE FINGER.....	6
4 、 Enrollment and Verification Procedures	6
4.1. ENROLLING USER	7
4.2. TESTING AN ENROLLMENT.....	10
4.3. BACKUP ENROLLMENT	10
4.4. VERIFYING YOUR IDENTITY	10
4.4.1. Fingerprint verification	11
4.4.2. Password verification.....	11
4.4.3. ID&Fingerprint	11
4.5. HINTS FOR SUCESSFUL ENROLLMENTS	12
5 、 System options.....	12
5.1. SYSTEM OPTIONS	13
5.1.1. Date time	14
5.1.2. Language	14
5.1.3. Format.....	14
5.1.4. Advanced options	14
5.2 COMMUNICATION	
OPTION.....	16
5.3. LOG OPTION	17
5.4. ACCESS CONTROL FUNCTION OPTION.....	17
6 、 System information	19
7 、 Maintenance.....	19
8 、 User access control guide.....	20
8.1. FEATURES.....	20
8.2. BASICE CONCEPTS	21
8.3.1. Brief introduction.....	23
8.4. VERIFYING PROCEDURE	24
8.5. FUNCTION INTRODUCTION	24
8.5.1. Time period	24
8.5.2. Grouping function.....	26
8.5.3. User access control setting.....	27

8.5.4. Open door combination function.....	29
8.5.5. Lock control time.....	31
8.5.6. Door magnet delay	31
8.5.7. Door magnet switch	32
TECHNICAL SPECIFICATION	

Tables

Table2-1Match threshold description.....	5
Table3-1The reasons of enroll failed or poor quality.....	12
Table6-1Maintenance	19

1 Introduction

This chapter will explain the fingerprint device appearance in details.

Access Control



OK Confirm key: Confirm your present operation.

MENU Menu key: Press down the key for seconds to enter the management interface in the initial status.

C Cancel key: Cancel your present operation.

↑ Scrolling up key. Scrolling up the menu (Check IN)

↓ Scrolling down key. Scrolling down the menu(Check OUT)

⊙Power key: Control the power on and off

BELL Sound bell

Number: Press number 0---9

Access control is a simple, easy to use terminal machine, it could be installed in the sensitive area where access requires authority, innovative manner combine the advanced security fingerprint recognition with access control management, to achieve the high security management of access and exit control

2 User information

Please don't put the device in the direct sunlight, strong sunshine will influence the fingerprint collection, and cause the verification invalid.

Try not to use the device outside in summer, its working temperature is 0-40 degree, long-time use outside and the machine's fever will affect its normal work, it will get slower reaction, lower through rate. If the device has to be used outside, you are recommended to adopt a umbrella and heat output equipment to protect the machine.

3 Basic concepts

3.1 Basic concepts

This section contains definitions and descriptions of Fingerprint T&A Concepts including:

- User Enrollment
- User Verification
- Match Threshold Levels
- User ID Numbers
- Authority (Status) Levels
- The two most important Fingerprint T&A Functions are user enrollment and user verification.

3.1.1 User Enrollment

During user verification, the stored template is compared to the current fingerprint scan to confirm the user's identity. The user can verify on the time recorder that enroll his fingerprint. The enrollment process takes approximately 1 second. Up to ten fingers can be enrolled for the same ID number, so the user may have more verifications way.

Ideally, everyone finger on each hand should be enrolled so that if the user injures the enrolled finger had, an alternate finger template is available. While recording the finger type is optional, it is recommended such as left or right index, so that at verification the user is prompted as to which finger to use.

3.1.2 User Verification

Verification occurs when a user either enters an ID number or places a finger in the fingerprint sensor, then input password for comparison with the stored template.

3.1.3 Match Threshold Levels

The Match Threshold Levels establishes a balance between False Acceptance Rate (FAR)

and False Rejection Rate (FRR). FAR means how often a non-authorized user is falsely recognized and granted access to the system. FRR means how often an enrolled and authorized user, who should be granted access to the system, is denied on the basis that the system did not recognize him.

You can set match threshold levels on a per-use basis. For the fingerprint verification is difficult, you can adopt ID & Fingerprint verification (match one to one). Raising the threshold increases security, while lowering it increases throughput. The correct balance is essential.

For users with worn or damaged fingers, match threshold levels can (and should) be reduced. (Please refer to Table 2-1).

Different sites have different requirements.

- ☞ **Note:** FAR and FRR affect each other. Reduce FAR will rise FRR, same on the contrary. Default security value is **3**, default **1:1** value is **40**. **Table 2-1** is the settings to security value in different occasion

Table 3-1 Match threshold description

FRR	FAR	Threshold Levels	
		1 : N	1 : 1
High	Low	4	50
Middle	Middle	3	40
Low	High	2	30

3.1.4 User ID number

Before beginning fingerprint enrollment, a user is assigned User ID Number. This ID number is used to input the fingerprint template each time that verification is requested. ID numbers are normally input via the keypad, but they can be input through other means such as RF Card. (The premise is that equipment must be with an RF card reader)

3.1.5 Authority(status) Levels

Fingerprint T&A has four authority or status levels:

● **Users** are people whose identity must be verified, such to gain access to a facility or to have their attendance recorded.

● **Enrollers** are Users who are authorized to enroll new users or delete on to the system.

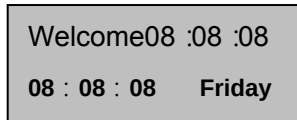
● **Managers** can do other operations, except set advanced option and enroll manager's authority.

● **Supervisors** are Users who access to all functions and change all setup in the system.

Note: Without Manager and Supervisor status in the system, the Enroller will enroll them. And if there isn't a Supervisor in the system, the Manager will enroll him.

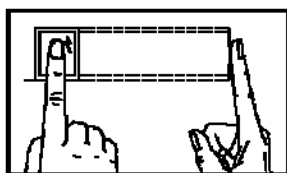
3.1.6 Start up window

Click the power button and the screen showed on, a display of a window was called the Start-up window. The following appears:



3.2 How to place the finger

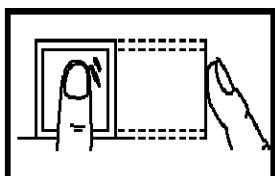
1) The correct way is:



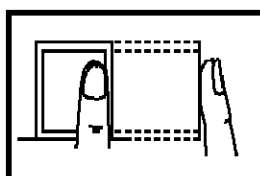
Place a finger flat on the sensor surface
Place it to the center of the sensor surface

2) The wrong way is:

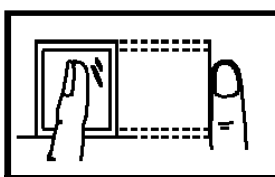
Vertical



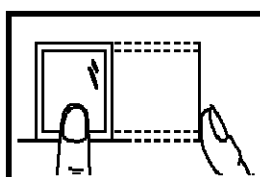
Offset



Inclined



Lower



Note: Please adopt the correct way to place the finger, our company is not responsible for the malfunctioning results arising from improper pressing manners. Our company reserve ultimate the interpreting and modifying right for this.

4 Enrollment and Verification Procedures

This chapter describes how to enroll and verify users on the Fingerprint T&A verification system, moreover, it will introduce how to verify the consistent of the fingerprint enrollment..

The following topics are included:

- ✧ Enrolling User
- ✧ Testing an Enrollment
- ✧ Enrolling a User with the Display Finger Option On
- ✧ Verifying Your Identity
- ✧ Hints for Successful Enrollments

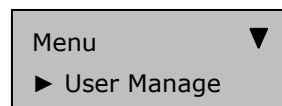
Note: You must have Enroller, Manager, or Supervisor status to enroll users. For information on status levels, see "Authority (Status) Levels" on 2.1.5"

4.1 Enrolling User

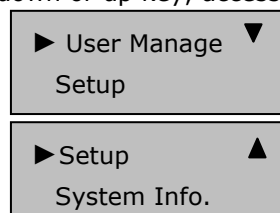
If this is the first enrollment in a new or empty system, everyone will become an enroller. If there is a manager in the system, you will have Manager status to enroll a new user. There are three ways of enrollment, fingerprint enrollment, password enrollment, fingerprint & password are suitable for three different crowds. Fingerprint enrollment suited the most people who have better quality fingerprints; Fingerprint & Password suited few people who enrolling successful, but verification was difficult; Password enrollment suited 1% people who verification unsuccessful.

To start the enrollment process, identify yourself firstly—press [Menu], enter your ID number or fingerprint, and then verify your identity.

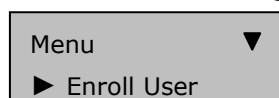
☞ Note: If there is no manager in the machine you should not verify your ID.. The following message appears on the display:



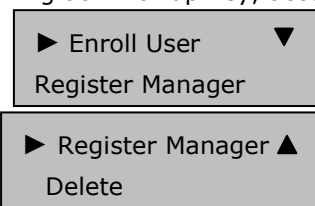
Press scrolling down or up key, access the right option:



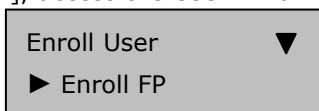
Press [OK], access the User Manager, the following message display:



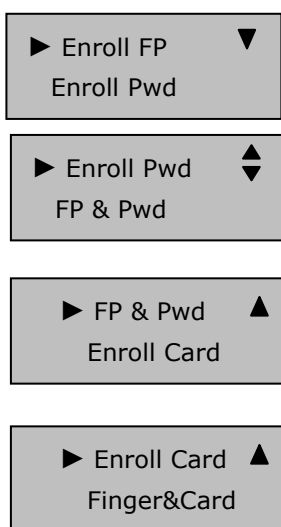
Press scrolling down or up key, access the right option:



Press [OK], access the User Enrollment, the following message display:

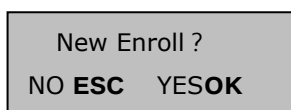


Press scrolling down or up key, access the right option:

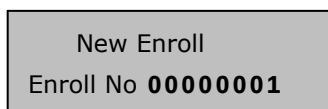


1. Fingerprint Enrollment

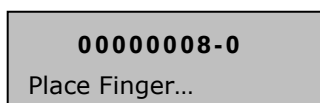
1) Access the Enroll Fingerprint to press [OK], the following display:



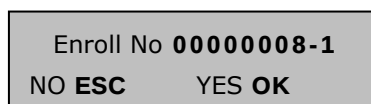
2) Press [OK], the following appears:



3) Input the enrolling number (the range is 1 to 999999999), press [OK], the following appears:



4) If the test is successful three times in a row, the following appears:



Note : **00000008-1**

The last number 1 means the first fingerprint.

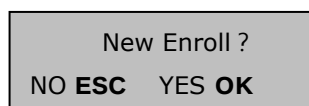
Press [OK], the previous message continues to be displayed while the template is created.

If your identity cannot be verified, you are prompted to try again. And you must restart the verification procedure (step 3).

2. Password Enrollment

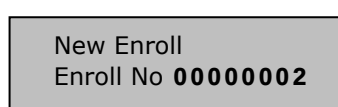
Access the password enrollment, press [OK], the following appears:

1) Press [OK], the following appears:



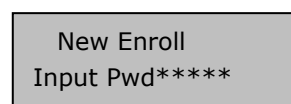
New Enroll ?
NO **ESC** YES **OK**

2) Press [OK], the following appears:



New Enroll
Enroll No **00000002**

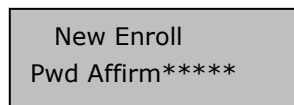
3) Input the enrolling number (the range is 1 to 99999999), press [OK], the following appears:



New Enroll
Input Pwd*****

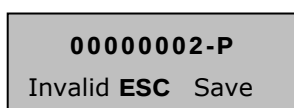
Note: The password range is from 1 to 8

4) Input the password, press [OK], the following appears:



New Enroll
Pwd Affirm*****

5) Input the password again, press [OK], the following appears:



00000002-P
Invalid **ESC** Save

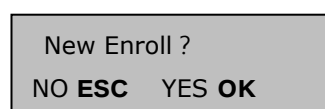
Note: 00008-P

The last letter P means
password.

Press [OK] to save the enrolled data, then finish once password enrollment procedure.

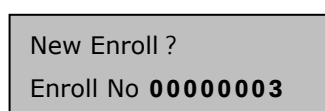
3 · Fingerprint & Password

1) Access the Fingerprint& Password, press [OK], the following appears:



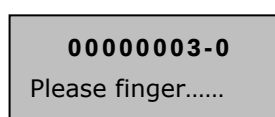
New Enroll ?
NO **ESC** YES **OK**

2) Press [OK], the following appears:



New Enroll ?
Enroll No **00000003**

3) Input the password (range is 1 to 99999999), press [OK], the following appears:



00000003-0
Please finger.....

4) If the test is successful three times in a row, the following appears:

New Enroll
Input Pwd:*****

5) Input your password, the following appears:

New Enroll
Pwd Affirm*****

6) Input the password again, press [OK], the following appears:

00000003- 1P
Invalid **ESC** Save**OK**

Note: **00000003-1P**
The second from the last 1
means one fingerprint

Press [OK], the previous message continues to be displayed while the template is created.

4.2 Testing an Enrollment

Ask the users to place their finger to perform test verification. If the test was successful, you will adopt Fingerprint enrollment. If the poor quality of fingerprint, or to re-enroll or change the finger, if that doesn't work, you will be recommended to use Fingerprint & Password.

4.3 Backup Enrollment

In the new enrollment interface, Press [OK] to cancel new enrollment, Press [ESC] to the Display finger Option On, the following appears:

New Enroll
NO **ESC** YES**OK**

Backup Enroll
Enroll No **00000003**

Following operation steps is consistent with new enroll, only the „New Enroll“ changes to be „Backup Enroll“

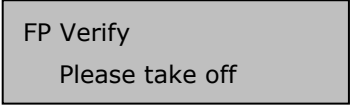
 Note: Where system memory permits, it is advisable to have at least two fingers enrolled for a long-term user.

4.4 Verifying Your Identity

The workers used fingerprint T&A to verify their identity. The system defaults to three ways of verification: Fingerprint verification, Password verification, ID number & Fingerprint. The following appears:

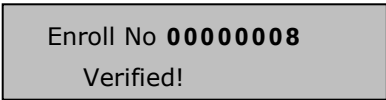
4.4.1 Fingerprint verification

Place the finger on the initial interface, the following appears:



FP Verify
Please take off

It continues for about 0.5 seconds. If the test is successful, it will say "Thank you", the following appears:



Enroll No **00000008**
Verified!

If your identity cannot be verified, you are prompted to try again, the following appears:

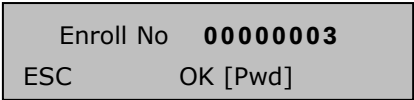


FP Verify
Place Finger Agn !

It continues for about 0.5 seconds, return to the start-up window.

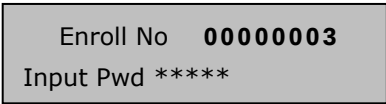
4.4.2 Password Verification

To start the enrollment process, enter your ID number, the following appears:



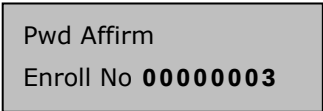
Enroll No **00000003**
ESC OK [Pwd]

Press [OK], the following appears:



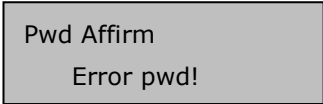
Enroll No **00000003**
Input Pwd *****

Input correct password, press [OK], the following appears:



Pwd Affirm
Enroll No **00000003**

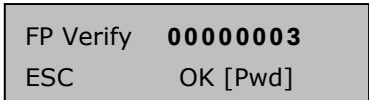
If the password cannot be verified, the following appears:



Pwd Affirm
Error pwd!

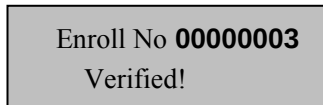
4.4.3 ID number & Fingerprint

To start the enrollment process, enter your ID number, the following appears:



FP Verify **00000003**
ESC OK [Pwd]

Press finger on the interface, if your identity is verified, the following appears:



4.5 Hints for Successful Enrollments

The fingerprint is of good quality, the verification speed will fast; otherwise, it will verify slowly or occur to FRR.

In order to improve the fingerprint verification quality, the following hint appears:

Table4-1 Enrolled unsuccessful or the fingerprint of poor quality

Dry or dirty fingerprints

If the fingerprint is dry, you can adopt the way of wetting up the finger, such as breathing at the finger's tip. Pls. keep your finger in clean.

Have no enough to bring pressure

The user should place a finger firmly and squarely on the sensor surface.

How to select the finger?

Recommend to left or right index or middle finger.

Use fingerprints of good quality, without worn or hurt.

The user usually selects the forefinger, if it was of a poor quality, you will be recommended to middle finger or ring finger.

If the user's finger is less, you will select the thumb. If the user need more backup fingerprint, you may select ring finger.

How to place the finger?

Place your finger firmly and touch the sensor surface. The fingerprint does not touch upright on the sensor surface. Do not touch the finger too fast; Do not move the finger on the sensor surface.

The influence of the fingerprints change

For users with worn or damaged fingers, the identification can be influenced.

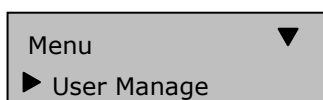
If the finger quality was poor, you have to re-enroll, or select password verification.

Others

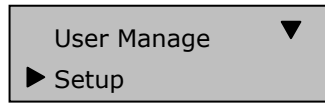
However, few people's fingerprint quality is too poor to verify the fingerprint in gear. Please use ID & fingerprint verification, and the Match Threshold Levels can be reduced or use password verification.

5 System Options

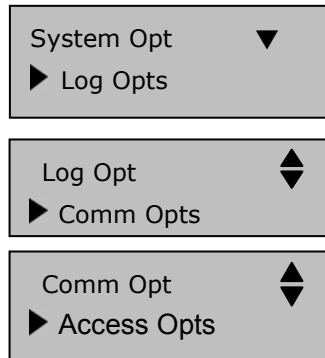
Press the [Menu], and then verify your identity. The following appears:



Press scrolling down or up key, access the right option:



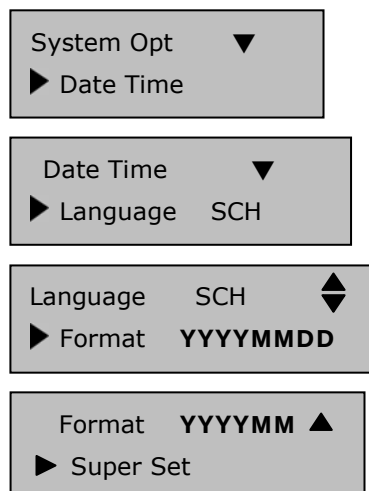
Access next operation step, press [MENU], the following appears:



The following topics included: System Options, Record Setup, Communication Options, Log Options.

5.1 System Options

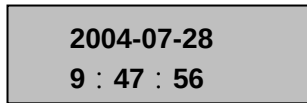
Access System Opt, the following appears:



The system setup facility covers four areas: Date Time, Language, Format and Advanced Options.

5.1.1 Date Time

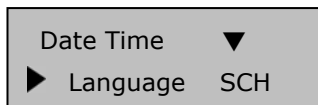
Access Date Time, the following appears:



To change the date, press  and , then input the correct date and time, press [OK]

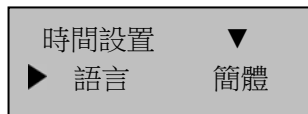
5.1.2 Language

Access Language, press [OK], it defaults to SCH.

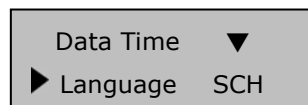


Access Language, press [OK], it defaults to SCH. Press the key to select another language. Fingerprint T&A comes with three built-in languages, simplified Chinese, traditional Chinese and English for all displays.

Select the language that you will use, press [OK], and then Press [ESC] to return the Options. The system will ask for confirmation of the displayed language. Press [OK] to accept the language displayed.



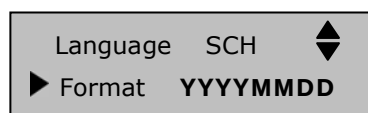
Traditional Chinese



ENG

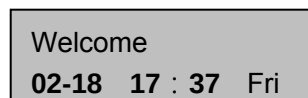
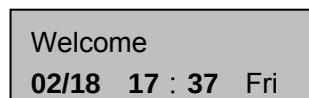
5.1.3 Format

Press Up or Down key to select "Format", press "OK" to process modification.



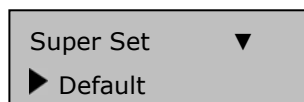
There are four kinds of formats: **YYYYMMDDYY.MM.DDYY-MM-DDYY/MM/DD**. It's the modification on the initial interface.

i.e. : **YY/MM/DD** (left pic.) 、 **YY-MM-DD** (right pic)

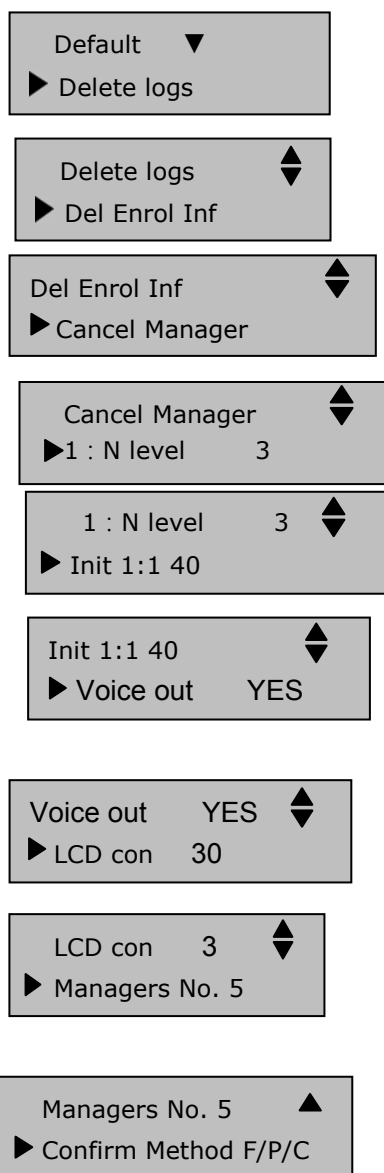


5.1.4 Advanced Options

Access Advanced Option, press [OK], the following appears:



Press and to scroll up or down the screen to select the option.



Press the scroll up or down key to select the option.

Restore Default: restore all setup information to default.

Clear all Data: delete all enrolling fingerprints and logs.

Delete Logs: delete all logs of flash disk.

Clear Managers' rights: change the managers' rights into ordinary users' rights.

1:N Security settings: Setup owner's security value

1:1 Settings: Setup single people's security value

Sound: whether use to phonic hint or not.

LCD Backlight: Setting the contrast of the screen

Manger No.: Setting the whole number of the managers

Confirm Method: Setting T&A access control method, F/P/C is fingerprint, password or card. F+P is fingerprint and password, F+C is fingerprint and Card. So that could pass verification

5.2 Log Options

Access Log Opt, the following message appears:

Log Opt	▼
▶ Alm Man Log	10

Alm Man Log	▼
▶ Alm Att Log	10

Alm Att Log	▲
▶ Repeat Confirm	0

Alarm Management Log: When the management logs reaches set numerical value, it will automatically sound a warning that logs were full.

Alarm Attendance Log: when the remainder log capacity reached set numerical value, it will automatically sound a warning that logs were full.

Repeat Confirm time: During the setup time (unit:mins), if there is already someone's attendance logs, the second attendance result won't be stored.

5.3 Communication Options

Access Communication Option, the following message appears:

Comm device	▼
▶ Dev Num	1

Dev Num	▼
▶ Baud Rate	38400

Baud Rate	38400	◆
Port	5005	

Port	5005	◆
Comm PWD	0	

Comm PWD	0	◆
IP address		

IP address	◆
Mask	

Mask	◆
Default Gateway	

Default Gateway	▲
Server IP Address	

Server IP Address ▲
▶ Server Port 5005

Server Port 5005▲
▶ Transmitting NO

Device Number: machine ID, the range is 1 to 255.

Baud Rate: there are three options, 9600, 38400, 115200; the speed of low communication is stable, recommend to use RS485.

Port: TCP's communication port, default to be 5005.

Communication Password: Its required in TCP communication.

IP address: it defaults to IP address is 192.168.1.204, could also get modified according to demands.

Mask: It could be setup to own internet.

Default Gateway: It could be setup to own internet.

Server IP address: Which installed access control management software, could automatically upload records from the access control device to your computer.

Server Port: Install with access control management software to gather TCP/I port of access control device.

Transmitting: Whether the settings automatically upload record to the management software.



Note: RS485 communication using the outdoor wiring, the need to install anti-

lightning devices.

5.4 Access control function option set

Access control option, the following message appears:

Function Opt ▼
▶ Define TP

Define TP ▼
▶ User access opts

User access opts ▼▲
▶ GRP TP Define

GRP TP Define ▼▲
▶ Unlock Group

Unlock Group		◆
▶ Lock control time	5	

Lock control time	5	◆
▶ Door Magnet Setting	10	

Door Magnet Type	NO
Lock delay	10

Door sensor set	▼
▶ Alarm Time	1

▶ Alarm Time	1	▼
Turn off alarm		

Turn off alarm	◆
Open Lock Mode	

G Mode	No	◆
Users	2	

There are nine main functions in the access control option

- 1) Define TP:** Define the time when the device could be unlocked everyday in a week
- 2) User Acc Opts:** The setup of unlock time and combination
- 3) GRP TP Define:** The group and configure the group Time period
- 4) Unlock Group:** The definition is the definition of combination lock that can unlock different combinations, each combination composed by different groups;
- 5) Lock control time:** The fingerprint control the time to turn on the lock.
- 6) Lock delay:** That means the device start to alarm after a certain time when the door is opened
- 7) Door magnet type:** includes three types: No, Normally opened, Normally closed; No means not use door magnet switch, normally opened means it keeps opening after unlock, normally closed means it keeps closing after lock
- 8) Alarm Time:** It starts to alarm after illegal door open, or device removal
- 9) Turn off alarm:** After illegal door open or device removal, the alarm could be removed
- 10) Access Mode**
 - Use normal mode
 - Yes: start up the combination and confirm multi-user function
 - No: start up the multi-user function irrelevant to the combination and confirm it.
 - Confirm the user number
 - It means when "use general mode=No", the user number which need to be recognized
 - User number=1-5

If set up "use general mode=No", no matter which group, you only need to confirm some

users at the same time(the number of people could be setup in “user number confirm”) to unlock the device. If one people passed as several people, the lock couldn’t be unlocked.

If the user identification failed 5 times, previous successful identification will become invalid, it requires reidentification.

If the user identification failed more than 10 sec., previous successful identification will become invalid, it requires reidentification too.

In this menu, each function could be defined, you may check **user access control guide** for the detailed operation.

6 System Information

Access [Menu] to Sys Info, press [OK], the following appears:

Sys Info	▼
► User Cnt	206
Man Cnt	5
FP Cnt	211
Card Cnt	1 ▲▼
Pass Cnt	1
G-Log	987 ▲▼
S-Log	3
Management Records	▲▼
Device info	

It displays user numbers, password numbers, fingerprint numbers, manager and log numbers in the screen. You can see some information such as the remainder log capacity in the memory, the device capacity, the defaults date, machine number and OEM’s information are in the system.

7 Maintenance

1 Cleaning

From time to time, the optical platen, the keypad and display window require cleaning. Since working environments differ, it is not possible to define when cleaning should be performed. Following is guide:

Table 7-1 Maintenance

Item	Cleaning frequency
Keypad and display window	Cleaning when visibly dirty and hard to read. See cleaning the keypad and display below.
Optical platen	Do not over clean. The platen is designed to work under greasy or dirty conditions.
	However, do clean if the platen is obstructed or people are reporting deteriorating performance. See cleaning the Optical Platen below.

2 Cleaning the Keypad and Display

To clean the keypad and display, use the same cleaning products as for the platen, and wipe dry.

3 Cleaning the Optical Platen Clean the optical platen as follows:

- (1) If dusty or gritty, first blow on the platen to clean off any loose particles.
- (2) Spray on a light coating of a window cleaning product or other similar neutral detergent.

Warning: Do not use any other cleaner or the platen may become damaged.

- (3) Using a non-abrasive and lint-free cloth, pat dry. Be careful not to scratch the platen. If lint particles adhere to the platen surface, remove them off when the platen is dry.

8 User access control function Guide

8.1 Features

- ◇ Anytime grouping access control
- ◇ Access control and Attendance could be managed
- ◇ Inner integrated single door controller, simple installation.
- ◇ Red and green lights on the panel
- ◇ Device removal alarm function
- ◇ Standard access control steel installation
- ◇ Internet management software, monitor several machines
- ◇ U disk data downloading is optional
- ◇ Wiegand 26 standard output (optional), compatible to other access controller
- ◇ Adopt WFS6.00 High-speed algorithm

8.2 Basic Concepts

Access control basic concepts:

- ◇ **Function setup:** enroll the unlock Time period and combination
- ◇ **Time period definition:** define the time when the device could be unlocked everyday in a week
- ◇ **Grouping:** divide the enrolled users to several groups make it manageable

- ◇ **Valid Time period:** the user verification needed time
- ◇ **Group Time period:** setup the group open lock time
- ◇ **User Time period:** setup the user open lock time
- ◇ **Open lock combination:** Define different can unlock combination, each combination is made up of different groups.
- ◇ **Lock control time** The fingerprint control the time to turn on the lock.
- ◇ **Door magnet delay** That means the device start to alarm after a certain time when the door is opened
- ◇ **Door magnet type** includes three types: No, Normally opened, Normally closed; No means not use door magnet switch, normally opened means it keeps opening after unlock, normally closed means it keeps closing after lock
- ◇ **Use normal mode**
 - Yes: start up the combination and confirm multi-user function
 - No: start up the multi-user function irrelevant to the combination and confirm it.

Confirm the user quantity

It means when "use general mode=No", the user number which need to be recognized

User quantity =1-5

If set up "use general mode=No", no matter which group, you only need to confirm some users at the same time(the number of people could be setup in "user number confirm") to unlock the device. If one people passed as several people, the lock couldn't be unlocked.

If the user identification failed 5 times, previous successful identification will become invalid, it requires reidentification. If the user identification failed more than 10 sec., previous successful identification will become invalid, it requires reidentification too.

In this menu, each function could be defined, you may check user access control guide for the detailed operation.

8.3 Access control function

Pres "Up/Down" to enter "Setup", press" OK" to access setup menu

Pres "Up/Down" to enter "Access control setup", press "Ok" to access function menu, the followings message displays:

Function Opt ▶ Define TP	▼
Define TP ▶ User Acc Opts	▼
User Acc Opts ▶ GRP TP Define	◄
GRP TP Define ▶ Unlock Group	◄

Unlock Group		◆
▶ Lock Control Time	5	
▶ Lock Control Time	5	◆
▶ Door Sensor Set	10	
Door Maganet Type	NO	
Lock Delay	1	
Door Sensor Set		▼
▶ Alarm time 1		
▶ Alarm time 1		▼
Turn off alarm		
Turn off alarm		◆
Open Lock Mode		
Access Mode	No	◆
Confirm User No.	2	

There are nine main functions in the access control option

- 1) Define TP:** Define the time when the device could be unlocked everyday in a week
- 2) User Acc Opts:** The setup of unlock time and combination
- 3) GRP TP Define:** The group and configure the group Time period
- 4) Unlock Group:** The definition is the definition of combination lock that can unlock different combinations, each combination composed by different groups;
- 5) Lock control time:** The fingerprint control the time to turn on the lock.
- 6) Lock Delay:** That means the device start to alarm after a certain time when the door is opened
- 7) Door magnet type:** includes three types: No, Normally opened, Normally closed; No means not use door magnet switch, normally opened means it keeps opening after unlock, normally closed means it keeps closing after lock
- 8) Alarm time:** It starts to alarm after illegal door open, or device removal
- 9) Turn off alarm:** After illegal door open or device removal, the alarm could be removed
- 10) Unlock Mode**
 - Use normal mode
 - Yes: start up the combination and confirm multi-user function
 - No: start up the multi-user function irrelevant to the combination and confirm it.
 - Confirm the user number
 - It means when "use general mode=No", the user number which need to be recognized
 - User quantity=1-5

If set up "use general mode=No", no matter which group, you only need to confirm some

users at the same time(the number of people could be setup in “user number confirm”) to unlock the device. If one people passed as several people, the lock couldn't be unlocked.

If the user identification failed 5 times, previous successful identification will become invalid, it requires reidentification.

If the user identification failed more than 10 sec., previous successful identification will become invalid, it requires reidentification too.

In this menu, each function could be defined, you may check user access control guide for the detailed operation.

8.3.1 Access control function description

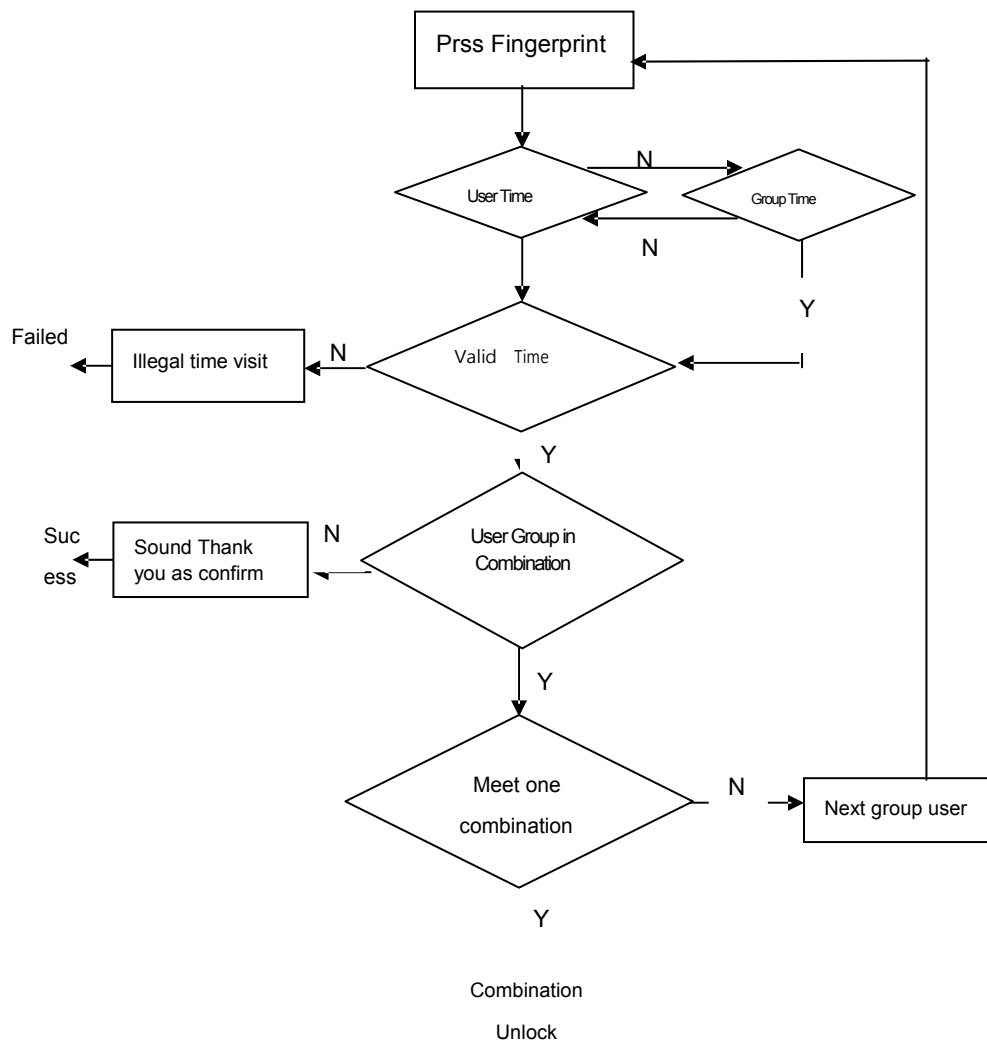
Each user setup is combined of identity group, use group Time period and user Time period. Grouping is dispatch user to certain group, i.e. group 1, group 2 etc.; group Time period could select three already setup Time periods at the most, those three Time period relation is “or”(meet anyone of the three is okay), user Time period could also select three already setup Time period too, their relation is “or” too; check “ **user access control function setup**” for the relation between group Time period and user Time period.

How the enrolled user open the lock

- 1 The group the user enrolled in should in the unlock combination(also could be in a unlock combination with other groups, but needs work together to open the lock)
2. The current open lock time should be in any valid time of user Time period

The system default new enrolled user as the first group, the grouping combination defaults to be the first group, default Time period is “1”, In the case of group 1 and Time period 1 are in factory status, new enroll user is default open lock status(if the user changed the related setup, the system will make according modification). If grouping combination setup doesn't have the user enrolled group, the user could only do attendance but open lock.

8.4 Access Control Verification Procedure



8.5 Function Introduction

8.5.1 Time Period Definition

Time period is the minimum unit in the access control setup. Whole system could define 50 time period at the most. Each time period defines 7 time periods namely 1 week, 24hours valid time period every day. Each user could setup 3 time period at the most, their relation is "or", the verification time could be valid if it meets any one of them. Each time format: **HH:MM-HH:MM, according to 24hours** Accurate to the minute.

End time less than start time (23:57- 23:56) means All-day ban, End time over than start time (00:00- 23:59) means the Time period is valid.

Note: System default time period code 1 is open whole day(new enrolled user open the lock)

- 1) Enter „Time period Definition“,the screen displays as following:

Time period Definition ▲	
▶ Time period Code	1

Press „OK“to access Time period 1 settings,the screen will display as following:

Time period 1 Definition ▼	
▶ SUN	00:00-23:59
SUN 00:00-23:59 ▼	
▶ MON	00:00-23:59
MON 00:00-23:59 ◆	
▶ TUE	00:00-23:59
TUE 00:00-23:59 ◆	
▶ WED	00:00-23:59
WED 00:00-23:59 ◆	
▶ THU	00:00-23:59
THU 00:00-23:59 ◆	
▶ FRI	00:00-23:59
FRI 00:00-23:59 ▲	
▶ SAT	00:00-23:59

Above Time period 1 is factory status

- 2) For example :

Time period could re-identify Time period code „1“ as:

Sat. Sun, rest, not allowed to access.

Mon. To Fri, its the work time, is allowed to access

Work time: 08:30-18:00

Settings as following

Time period 1 definition ▼	
SUN	23:57-23:56
SUN 23:57-23:57 ▼	
▶ MON	08:30-18:00
MON 08:30-18:00 ◆	
▶ TUE	08:30-18:00

TUE	08:30-18:0	◆
▶ WED	08:30-18:00	

WED	08:30-18:0	◆
▶ THU	08:30-18:00	

THU	08:30-18:0	◆
▶ FRI	08:30-18:00	

FRI	08:30-18:0	▲
▶ SAT	23:57-23:56	

Analogy, could setup several Time period, according to demands, the whole system could setup 50 Time periods at the most.

8.5.2 Grouping Function Definition

Grouping function divide the users to groups, and different groups can be combined into different unlocking combination, in order to facilitate access of the group management. Such as these can combine a lot of unlocking combinations. The system defines five groups: group 1, group 2, group 3, group 4, group 5. New enrolled users belong to default group, but can be re-assigned to other various groups.

Group time: Setup the group open lock time, choose already setup Time period code number in the group Time period.

New enrolled user default group 1 Time period, but when the user to reset their respective groups, the users use corresponding default Time period, so need define each group default Time period first of all.

How to use group time period, see "**User Access Control feature set**"

 **Note:** Group 1 Time period is default Time period, choose Time period definition code „1“. (namely the new enrolled user could open the lock)

a) Access „group default Time period“, the screen will display as following: :

Group default Time period
▲

b) Press "OK" to access the setup of group 1 default Time period, there are three Time periods in the group default Time period, their relation is „or“

Group 1 default Time period	1
-----------------------------	---

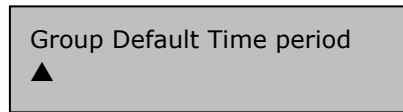
Time period 1	
Time period 2	8

Time period 2	
Time period 3	40

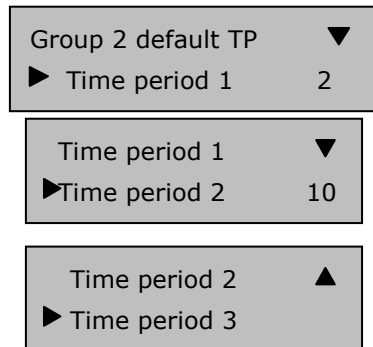
Group 1 is valid in Time period 1,8,40, could also select other Time period.

b) Access „group 2 default Time period“, the screen will display as following:

Group 2 Time period setup



Press „OK‘ to enter



Group 2 is valid in Time period 2, 20, 36, could also select other Time period.

Hereby, each group Time period could be defined according to demands, the system could define 5 groups Time period at the most,

8.5.3 User Access control Setup

User access control setup is based on user demands

1. Enter the menu to check certain user access control setup status

Includes: Affiliated group settings, group Time period and user Time period.

◇ Grouping: Divided the enrolled users to several groups, make the management convenient;

◇ Group Time period: Whether the user use affiliated group default Time period

◇ User Time period: Setup the user open lock time, choose the already setup Time period code

 Note: The relation between group Time period and user Time period „Yes“ or „No“ in the group Time period is only work to following „User Time period“

- 1) If the „group Time period“ is „Yes“, the user Time period will auto-pay value to affiliated group Time period code.
 - 2) If modified the user Time period, the „group Time period“ will auto be „No“
-

2. Setting an example

Setting the user 00000001、00000002 to be group 1 and group 2

A. Access 00001 interface, the screen will display as following

User Opt	
Att No. :	00000001

Press „ OK“ access „ Group Time period“ press Down or Up key to select „Yes“, the screen will display as following:

00000001 Access Control ▼	
▶ Affiliated group	1
▶ Group Time period	Yes

Affiliated group ▼	
▶ Group Time period	Yes
▶ Time period 1	1

Group Time period ▲▼	
▶ Time period 1	1

Time period 1 ▲▼	
▶ Time period 2	40

Time period 2 ▲	
▶ Time period 3	48

Code 00000001 user

User affiliated group is „1“, use group 1 Time period(user Time period code is the group Time period code)

00000001 user is available in Time period 1 ∼ 8 ∼ 48

B. Access 00002 setting interface, the screen will display as following:

User Opt	▲
Att No. :	00000002

Press OK to access

User Time period select code 1,20, :group Time period“ will auto be „No“

User00000002access control▼	
▶ Group	2
▶ Group Time period	Yes

Group ▼	
▶ Group Time period	No
▶ Time period 1	1

Group Time period ▲▼	
▶ Time period1	1

Time period 1 ▲▼	
▶ Time period 2	20

Time period 2 ▲	
▶ Time period 3	48

00000002 user:

User affiliated group is „2“, use user Time period not group Time period, namely it is valid in Time period 1, 20

If the user would like to use group Time period, select „Yes“, it will auto-pay value to group Time period code; otherwise, „group Time period will auto be „No“

8.5.4 Open lock combination function definition

Open lock combination is the direct display to control unlock, for example, if you would like to have all enrolled users couldn't open the lock, you may setup 10 open lock combination to be in vain.

Open lock combination definition is to define different combination which could open lock, each combination is made up of different groups. Open lock combination use group number directly, not consider user verification queue. For example „123“ means there should be one user each at least in group 1, group 2, group 3, together passed the verification to open the lock; „4“ means one user in separate group 4 could open the lock after verification. The system could define 10 open lock combinations Simultaneously at the most, if only one of them could pass the verification.

🔧 Note: System default open lock combination is „1“, (namely new enrolled user could open lock)

Press „Menu“ to access main interface;

Select „Setup“, press „OK“ to access setting menu

Select „Access control function setting“, press „OK“ to access menu

Select „Open lock combination definition“ access „open lock combination“, the screen will display as following:

Open Lock com. Defi.	
Combination1	1
Combination 1 1	
Combination 2	
Combination 2	
Combination 3	
Combination 3	
Combination 4	
Combination 9 ▲	
▶ Combination 10	

If didn't process settings(factory status), the system default group 1 as open lock

combination 1, others is in vain;if don't want all the user open the lock, setup 10 open lock combination as in vain.

1) For example:

Open lock com. Def. ▼	
▶ Combination 1	123
Combination 1 ▼	
▶ Combination 2	4
Combination 2 ◆	
▶ Combination 3	25
Combination 3 ◆	
▶ Combination 4	15

It shows from above combination settings:

123 is a combination

4 is a combination

25 is a combination

15 is a combination

Combination 1: If group 1,2,3, users are on scene, and at least one user pass the verification Time period is valid, the door would open

Combination 2: Only one user from group 4 on scene, the door will open

Combination 3: If group 2, 5 users are on scene, and at least one user pass the verification Time period is valid, the door would open

Combination 4: If group 1 5 users are on scene, and at least one user pass the verification Time period is valid, the door would open

❖ **Note:** The Time period which couldn't pass verification

- ✧ User Time period didn't select Time period code
- ✧ Group Time period didn't select Time period code (when using group Time period)
- ✧ User verification time is in neither Time period as the settings.
- ✧ Define the Time period as Prohibition

🗑 **Note:** If the setup Time period couldn't get passed.

❶ If agree above combination 2, appears the hints "illegal Time period visiting", the user couldn't open the lock, but can do attendance. (If there are some user could meet the open lock terms, group 2 could open lock too)

❷ If agree above combination 1, 3, 4, 5, it appears illegal Time period visiting", that combination user couldn't open the lock, but could do attendance.

2) For example: A bank vault needs three persona on scene at the same time, to open the vault door. Detailed settings as following:

Those three people separately belongs to group 2, 3, 5, has the open lock right at the same Time period. Select "Combination 1" press „OK" to access edit status, press number buttons, input 235, press ESC to exit and save the settings.

🗑 **Note:** If already set 1235 combination, couldn't set 12, 13, 14, 15, 23, 25, 35 combination anymore.

8.5.5 Lock Control Time

The system requires setup fingerprint device to control open lock time, set „0" as close lock control function, one number of units is 20ms, could setup as 254 at the most, i.e.: 5.08 seconds. Select this item and press „OK" to access settings, press number key to input corresponding number, press „ESC" to exit

8.5.6 Door Magnet Delay

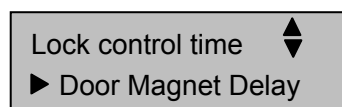
Door magnet delay is the device will alarm after certain time when the door opened, this period is door magnet delay

Select „Menu" access main interface

Select „Settings", press „OK" enter setting menu

Select „Access control function setting", press „OK" to access menu.

Press Up and Down key to access, select door magnet delay, it will appear as following:



Press OK, select the number, press up and down key to modify the door magnet delay time.

8.5.7 Door Magnet Switch

Door magnet type includes three types: No, Normally opened, Normally closed; No means not use door magnet switch, normally opened means it keeps opening after unlock, normally closed means it keeps closing after lock

Press "MENU" access main interface

Select "settings" , press "OK" to enter setting menu

Select "Access control function setting", press "OK" to enter menu;

Press Up and Down key and select door magnet switch, the following table shows:

Door magnet delay ▲	
▶ Door magnet switch	No

Press OK, select this item, press Up and Down key to shift the status

Door magnet delay ▲	
▶ Door magnet switch	
Normally open	

Door magnet delay ▲	
▶ Door magnet switch	
Normally closed	

Technical Specification:

	Specification
Capacity of Fingerprint	3000/10000
Capacity of record	60,000
Verification Method	1:1 or 1:N
Access control function	50 time, 5 groups, 10 open door combination, support multi fingerprint access, fingerprint or password verification method is optional, Way alarm output, device removal alarm output, etc.
Electric Lock control	3A/12VDC Relay Output
Attendance Function (optional)	support Attendance software
Other output	One group door magnet output, one group open door switch output. One group door bell output
Networking (optional)	TCP/IP、RS485、USB
Wiegand output	Wiegand26 output (optional)
Keyboard and display	LCD, 4*4 digital keyboard, blue back ++ light
Power	12V DC, Standby Current: 50mA, work current 400mA
Verify speed	=1 second
FRR	=1%
FAR	=0.0001%
Temperature operation	0°C - 45°C
Humidity operating	20%-80%
Language	Simplified Chinese, Traditional Chinese, English
OEM	Provide OEM job